

# Lecture Unit 1.5

## HPC languages

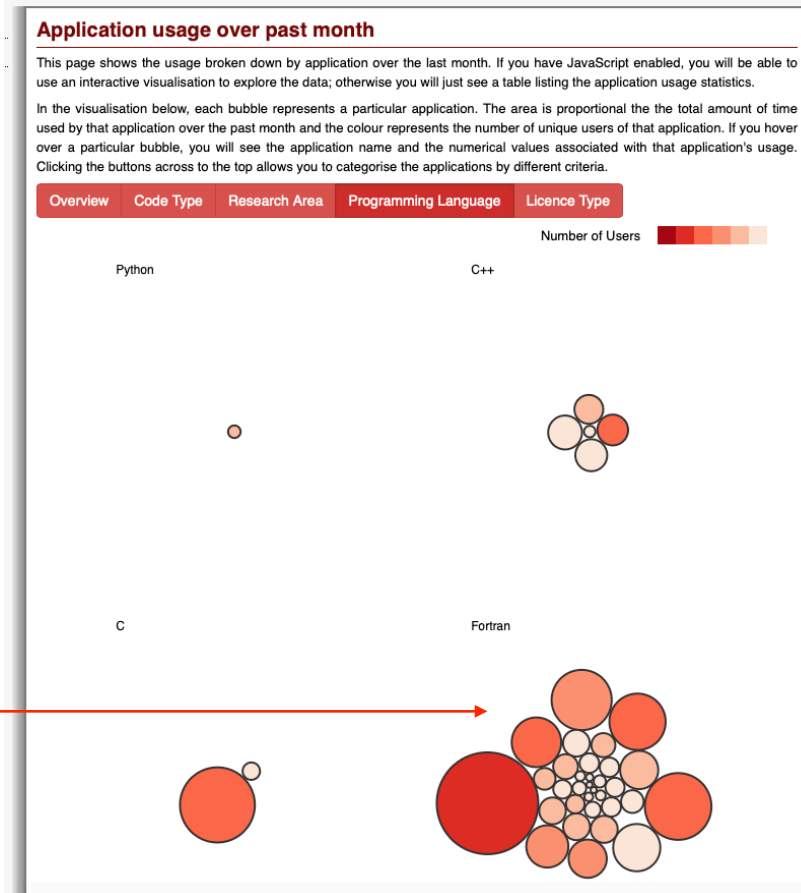
# HPC languages

- HPC applications typically use compiled languages to maximise performance
- Compilers parse the source code and generate an executable file
- Compilers can make optimisations at compile time which would be too expensive for an interpreter to make at run time
- C, C++ and Fortran are the most common languages used in HPC

# Applications on Archer

Applications on Archer (national HPC facility) broken down by programming language

Many large, legacy HPC codes and libraries are written in Fortran and would take enormous effort to rewrite



# C programming

- This module uses C for examples, workshops and assignments
- C is taught in the Exeter undergraduate programme in “The C Family” (ECM2424)
- If you need to refresh your knowledge of C there are a number of books in the library:
  - The “C Programming Language”, Brian W. Kernighan, Dennis M. Ritchie, available from the Library as an [e-book](#)
  - “Practical C Programming”, Steve Oualline, available from the Library as an [e-book](#)
  - See also [SAB](#) Appendix A (“Essential C”)

# Parallel programming

- Parallelisation is typically too complex for the compiler to handle automatically and needs to be added by the programmer
- Parallel programming is often based on extensions to existing serial languages
- OpenMP and MPI two widely used parallel programming methods in HPC which extend C, C++ and Fortran:
  - **OpenMP**: directives based parallelism for shared memory
  - **MPI**: Message Passing Interface for shared or distributed memory using function calls
- Specific parallel programming languages exist (e.g. Chapel and Julia) but are not as widely used at present

# Unit Summary

- HPC applications typically use compiled languages to maximise performance
- C, C++ and Fortran are the most common HPC languages
- Parallelisation normally needs to be added by the programmer
- OpenMP and MPI are used to parallelise programs written in C, C++ and Fortran
- See [SAB](#) Chapter 3.4